

TURDA, Romania

WMO: 151430

Lat: 46.5831N

Lon: 23.7911E

Elev: 1405

StdP: 13.97

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	14.5	5.3	7.6	14.6	8.7	9.1	17.7	24.0	23.6	19.2	27.2	2.2	10	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.7	87.4	67.6	84.5	66.9	81.8	66.2	70.6	82.9	69.2	81.2	67.8	78.9	4.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	102.5	76.1	65.0	97.4	74.4	63.8	93.3	72.9	35.3	83.2	34.1	81.8	32.9	79.1	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	14.6	12.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	27.1	33.9	42.2	51.6	60.7	67.4	68.4	69.4	61.0	51.0	41.2	31.0
	DBStd	16.19	7.43	8.02	8.15	8.83	6.32	5.46	5.19	5.70	7.63	7.21	7.64	6.59
	HDD50	2461	711	451	260	88	5	0	0	0	3	78	277	588
	HDD65	5776	1176	870	706	405	160	38	27	23	169	434	715	1053
	CDD50	2640	0	1	19	134	337	521	572	602	333	109	12	0
	CDD65	480	0	0	0	2	27	109	134	160	48	0	0	0
	CDH74	3671	0	0	0	40	211	812	951	1306	347	4	0	0
	CDH80	998	0	0	0	2	24	197	239	455	81	0	0	0
Wind	WSAvg	3.6	2.8	3.9	4.4	4.9	4.2	4.1	3.8	3.2	3.4	3.2	2.9	3.0
Precipitation	PrecAvg	21.1	0.9	0.7	1.0	1.7	2.5	3.5	3.3	1.9	2.1	1.7	1.3	1.2
	PrecMax	28.7	1.7	1.8	2.8	3.0	4.2	7.5	5.4	3.6	5.6	3.5	2.8	2.1
	PrecMin	13.6	0.3	0.3	0.2	0.2	0.8	0.7	0.6	0.1	0.2	0.0	0.0	0.2
	PrecStd	4.0	0.4	0.4	0.7	0.7	0.9	1.4	1.3	1.0	1.2	0.9	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	56.9	70.0	78.3	83.3	87.7	89.7	92.1	86.8	73.7	62.2	49.5
		MCWB	41.2	47.6	52.1	58.7	62.0	69.4	69.7	69.2	64.0	58.6	53.8	45.6
	2%	DB	40.6	51.6	63.6	74.1	79.3	85.0	85.7	88.3	82.0	70.5	57.7	44.7
		MCWB	38.4	45.7	49.4	57.1	62.2	68.2	68.6	66.8	63.8	57.4	51.3	42.5
	5%	DB	38.3	48.1	58.2	70.0	76.1	82.1	82.3	85.0	77.8	66.9	54.9	41.9
		MCWB	36.3	43.6	47.7	55.5	62.1	67.1	67.3	66.7	62.0	56.1	49.7	39.8
	10%	DB	36.0	44.7	53.7	65.4	72.4	79.0	79.4	81.5	73.8	62.5	52.1	39.1
		MCWB	34.5	40.8	44.8	53.4	59.6	65.9	66.4	66.4	61.2	54.4	47.8	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	49.9	53.4	62.2	66.6	71.8	72.0	71.6	68.4	61.1	54.8	46.9
		MCDB	45.5	55.2	66.2	72.3	77.6	83.7	82.7	85.7	81.2	68.4	59.8	49.1
	2%	WB	38.9	46.5	50.7	59.4	64.3	70.2	70.5	70.1	66.0	58.8	52.7	42.8
		MCDB	40.4	50.8	61.0	70.7	74.6	82.2	81.9	83.9	76.5	67.5	57.2	44.8
	5%	WB	36.6	43.9	48.1	57.0	62.7	68.5	69.0	68.5	64.1	56.9	50.4	40.1
		MCDB	38.0	47.8	57.1	68.4	73.2	79.8	80.0	81.1	74.0	65.5	54.6	41.8
	10%	WB	34.7	41.4	45.8	54.8	61.1	66.8	67.5	67.1	62.3	54.8	48.5	37.6
		MCDB	35.8	44.8	52.7	63.7	70.2	76.8	77.3	78.6	71.6	61.9	52.4	38.9
Mean Daily Temperature Range	5% DB	MDBR	9.1	11.8	15.8	18.5	17.8	18.5	18.7	19.7	17.5	16.4	11.1	7.6
		MCDBR	9.9	16.0	22.6	24.9	23.0	23.1	22.6	25.1	23.7	23.7	14.8	11.5
		MCWBR	8.4	10.8	12.2	11.6	10.2	9.5	9.2	8.7	8.8	11.7	8.9	9.3
	5% WB	MCDBR	9.4	15.1	21.2	22.6	21.1	22.0	20.7	22.6	20.9	21.4	14.1	11.5
		MCWBR	8.2	10.5	11.7	11.4	10.2	9.9	9.3	9.1	8.9	11.1	8.7	9.4
Clear-Sky Solar Irradiance	taub	0.306	0.332	0.371	0.416	0.402	0.399	0.417	0.422	0.384	0.357	0.333	0.308	
	taud	2.358	2.322	2.271	2.183	2.278	2.311	2.265	2.250	2.329	2.391	2.409	2.386	
	Ebn at Noon	247	264	269	266	274	275	268	261	261	252	235	233	
	Edh at Noon	23	30	36	43	41	40	41	40	34	28	23	21	
All-Sky Solar Radiation	RadAvg	387	645	1008	1379	1677	1840	1808	1667	1184	809	460	300	
	RadStd	54	80	138	140	159	153	158	156	135	98	62	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	+1.30	N/A	N/A	+1.69	N/A	N/A	-261	-374	+162	+43

Nomenclature: See separate page